Spectral Clustering on
Stochastic Block Models (SBM)

Understanding Community Detection Through Synthetic Networks

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1. Overview

This report explains how Spectral Clustering can be used to detect communities (groups) inside a network, using Stochastic Block Models (SBMs) as test graphs. The notebook walks through the mathematics, runs three experiments with different levels of community strength, and measures how well the algorithm recovers the true groups.

Key Points

- Stochastic Block Models (SBM) – generate graphs with known community structure
- Graph Laplacian – a matrix that encodes connectivity information
- Eigenvalues & Eigenvectors – reveal hidden structure in the graph
- Eigengap Heuristic – a rule to estimate the number of communities
- Spectral Clustering – uses eigenstructure + KMeans to assign nodes to communities
- Adjusted Rand Index (ARI) – measures how accurately communities are recovered

2. Key Definitions

2.1 Stochastic Block Model (SBM)

A Stochastic Block Model is a way to generate a random graph that has a hidden community structure built in. You choose how many nodes belong to each group, and then set two probabilities:

- **Intra-community probability** — how likely two nodes in the same group are connected (should be high).
- **Inter-community probability** — how likely two nodes in different groups are connected (should be low).

Because the true groups are already known, the SBM is perfect for testing community detection algorithms — we can check exactly how well they performed.

2.2 Adjacency Matrix (A)

A square matrix where entry A[i][j] = 1 if nodes i and j are connected, and 0 otherwise. For a 16-node graph the shape is (16, 16).

2.3 Degree Matrix (D)

A diagonal matrix where each diagonal entry D[i][i] equals the number of connections (degree) of node i. Example: if node 0 has 7 connections, D[0][0] = 7. All off-diagonal entries are 0.

2.4 Graph Laplacian (L)

Calculated as L = D - A (Degree Matrix minus Adjacency Matrix). The Laplacian is one of the most important matrices in network science because its eigenvalues directly reveal the community structure of the graph.

2.5 Eigenvalues & Eigenvectors

When you multiply a special vector by a matrix and you get back the same vector (just scaled), that vector is called an eigenvector and the scaling factor is its eigenvalue. For a Graph Laplacian:

- The smallest eigenvalue is always 0 (or very close to 0 due to floating-point precision).
- The number of zero eigenvalues equals the number of disconnected components.
- Small eigenvalues carry information about large-scale structure (communities).

2.6 Eigengap Heuristic

After sorting eigenvalues from smallest to largest, the eigengap is the difference between consecutive eigenvalues. A large jump (gap) between eigenvalue k and eigenvalue k+1 suggests the graph has k communities. Gap 1 (from 0 to the next value) is always ignored because the zero eigenvalue is a fixed property of any Laplacian.

2.7 Fiedler Vector

The eigenvector that corresponds to the second-smallest eigenvalue. It is used for two-way partitioning: nodes where the entry is negative go into community 1, nodes where the entry is positive go into community 2.

2.8 Spectral Clustering

A two-step process: first, pick the top k eigenvectors of the Laplacian and stack them as columns to form a low-dimensional embedding of the graph. Second, run KMeans clustering on these rows to assign each node to one of k communities. This works because nodes in the same community end up close together in the eigenvector space.

2.9 Adjusted Rand Index (ARI)

A score between -1 and 1 that measures how similar the predicted community labels are to the true labels. ARI = 1.0 means perfect recovery. ARI = 0 means the result is no better than random. Negative values mean worse than random.

3. Experiment 1 — Strong Community Structure

Graph setup: 2 communities of 8 nodes each (16 nodes total). Intra-community probability = 0.90, Inter-community probability = 0.05. The communities are very clearly separated.

3.1 Graph Visualisation

The spring layout places well-connected nodes close together. In the first plot nodes are coloured uniformly; in the second they are coloured by their true community labels, clearly showing two dense clusters with very few links between them.

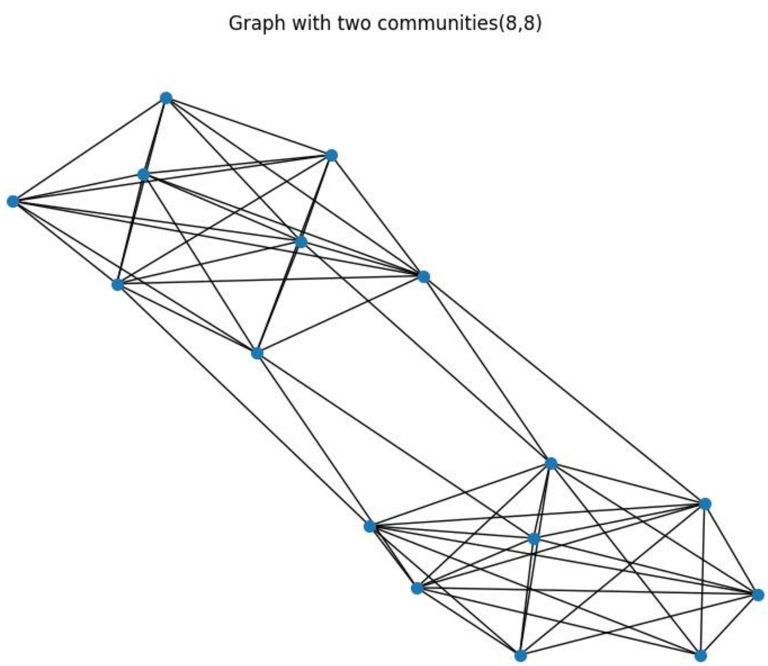

Fig 1a — Graph without community colouring

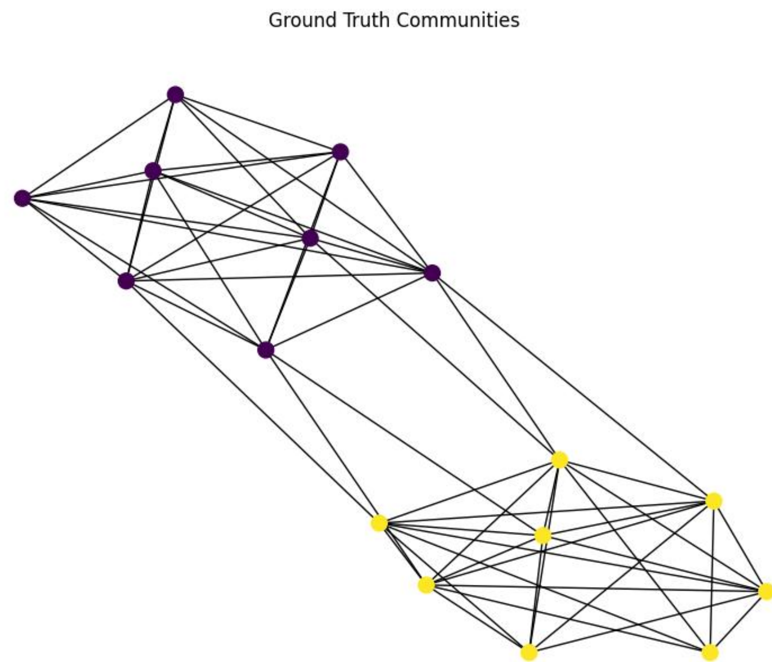

Fig 1b — Ground truth community colouring (yellow = community 0, purple = community 1)

3.2 Adjacency & Laplacian Matrices

The adjacency matrix A is a 16×16 binary matrix. The degree matrix D is computed by summing each row of A. The Laplacian L = D − A is also 16×16.

3.3 Eigenvalue Spectrum

After computing eigenvalues of L, the first value is essentially 0 (-3.55×10^{-15}, a floating-point artefact). The remaining eigenvalues increase. The eigenvalue spectrum plot shows all 16 values.

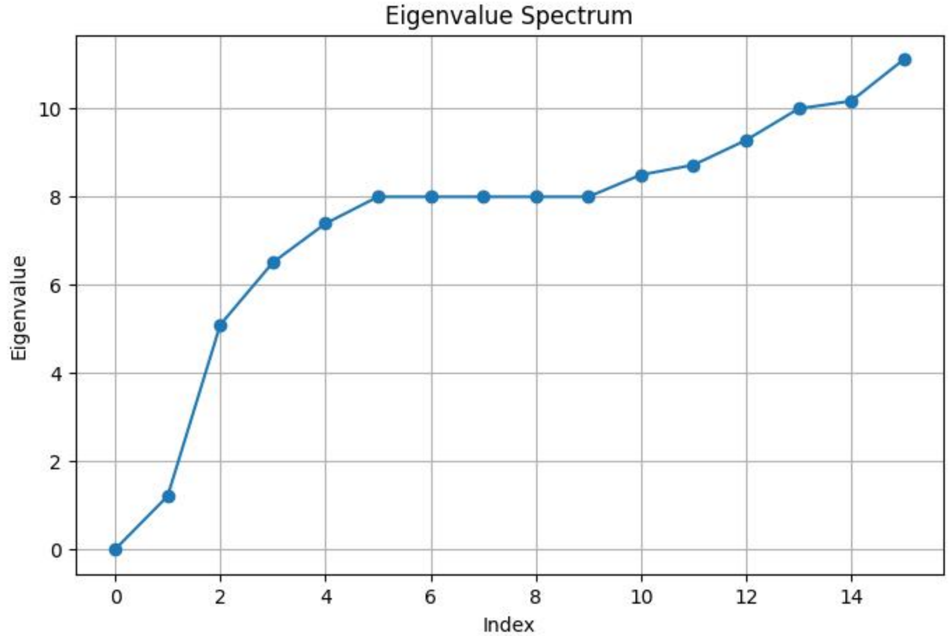

Fig 2 — Eigenvalue spectrum for the strong-structure graph

3.4 Eigengap Analysis

The gaps between consecutive eigenvalues are:

- Gap 1 = 1.2097 (ignored — always exists due to the zero eigenvalue)
- Gap 2 = 3.8849 ← Largest gap
- Gap 3 = 1.4136
- Gap 4 = 0.8843 …

The biggest gap falls after eigenvalue 2, so the eigengap heuristic correctly predicts k = 2 communities.

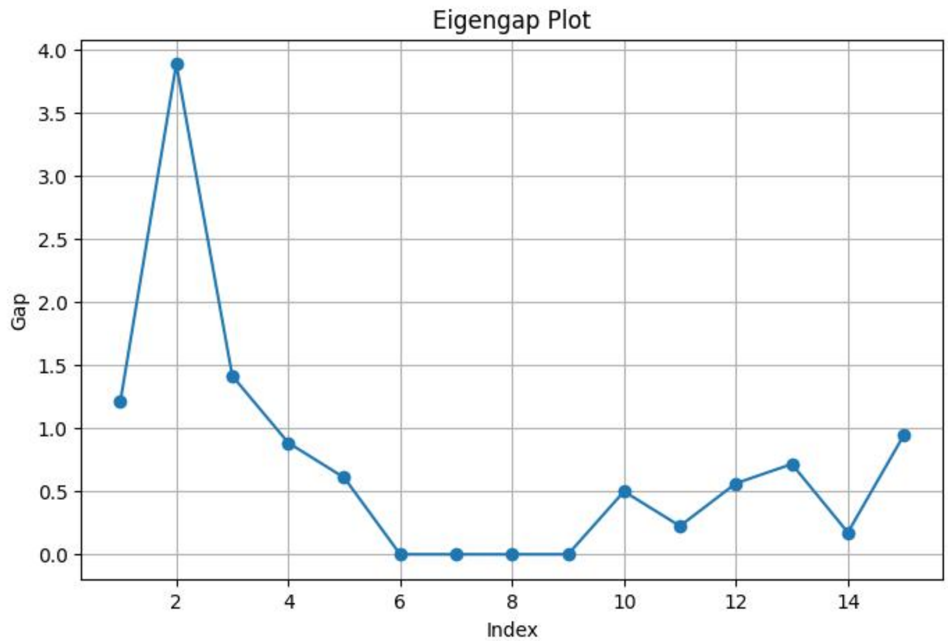

Fig 3 — Eigengap plot (Gap 2 is the tallest spike)

3.5 Fiedler Vector & Spectral Bipartitioning

The Fiedler vector cleanly divides into two groups: nodes 0–7 have negative values (community 0) and nodes 8–15 have positive values (community 1). The ARI is 1.0000 — perfect recovery.

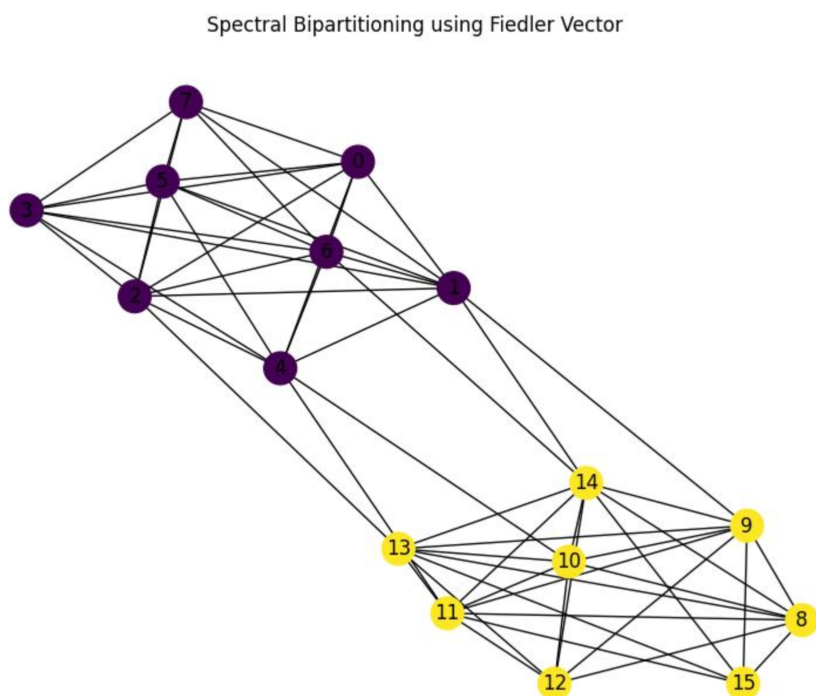

Fig 4 — Spectral bipartitioning using the Fiedler vector

3.6 Spectral Clustering (k = 2)

Running SpectralClustering with k = 2 also achieves ARI = 1.0000. The graph plot perfectly matches the ground truth.

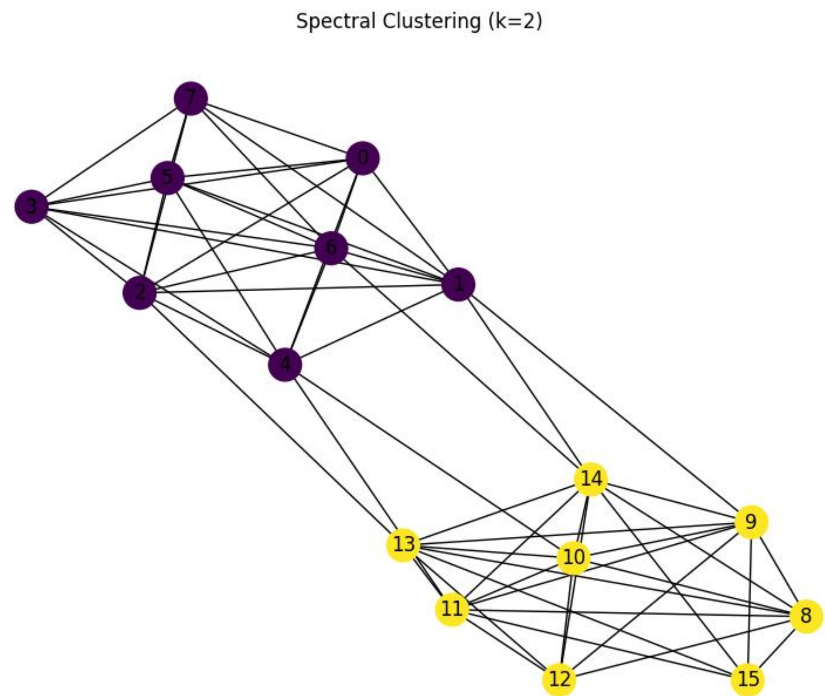

Fig 5 — Spectral clustering result (k = 2); perfect match with ground truth

Observation

- Both Fiedler bipartitioning and Spectral Clustering achieve ARI = 1.0000.
- The eigengap heuristic correctly identifies k = 2.
- When communities are very well separated, spectral methods work flawlessly.

4. Experiment 2 — Weak Community Structure

Graph setup: 3 communities of 7 nodes each (21 nodes total). Intra-community probability = 0.50, Inter-community probability = 0.35. The difference between inside-group and outside-group connections is very small — communities heavily overlap.

4.1 Graph Visualisation

The graph looks like a mixed ball of nodes. Even with the ground truth colouring, the three groups are hard to visually separate.

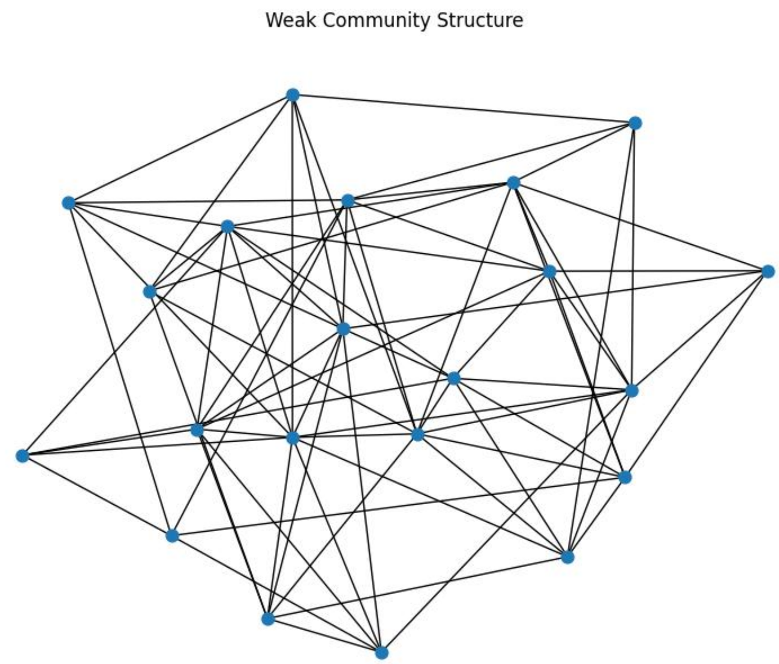

Fig 6a — Weak-structure graph (no colouring)

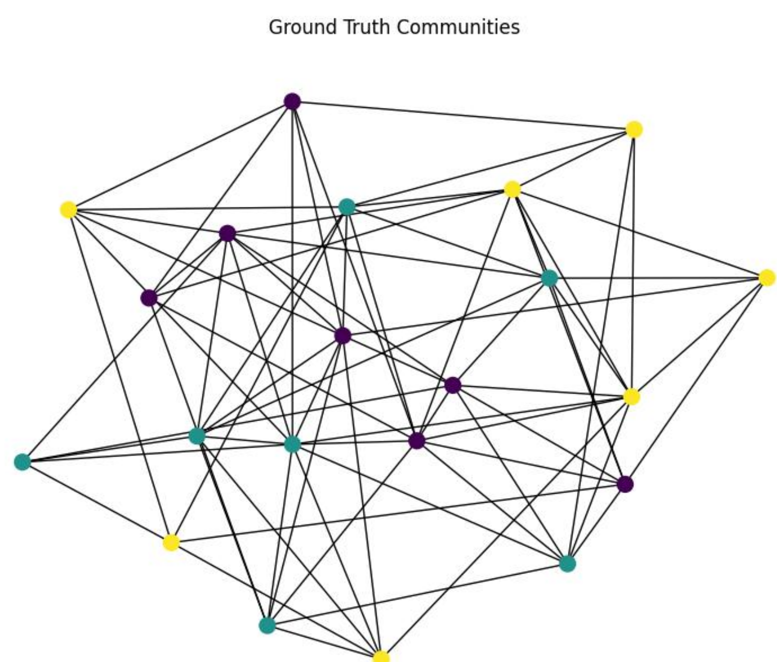

Fig 6b — Ground truth colouring; 3 planted communities, but heavily mixed

4.2 Eigengap Analysis

With weak structure the eigenvalue plot rises almost uniformly — there is no obvious gap. The heuristic incorrectly picks a large k because the biggest numerical gap appears late in the spectrum.

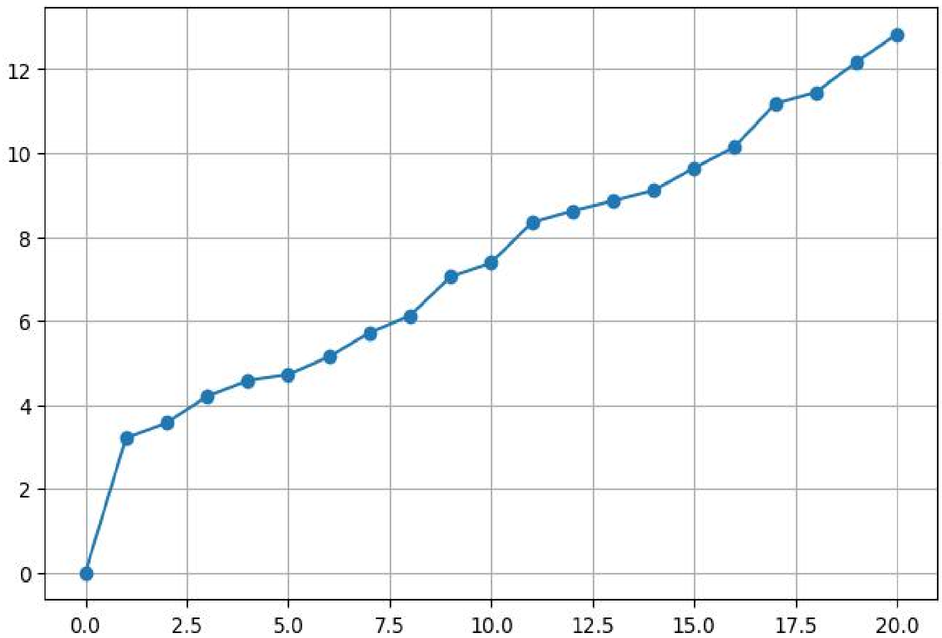

Fig 7 — Eigenvalue spectrum for weak-structure graph (no clear elbow)

4.3 Spectral Clustering Result

With an unreliable k estimate, spectral clustering assigns nodes almost randomly. ARI = −0.0154 (worse than random chance).

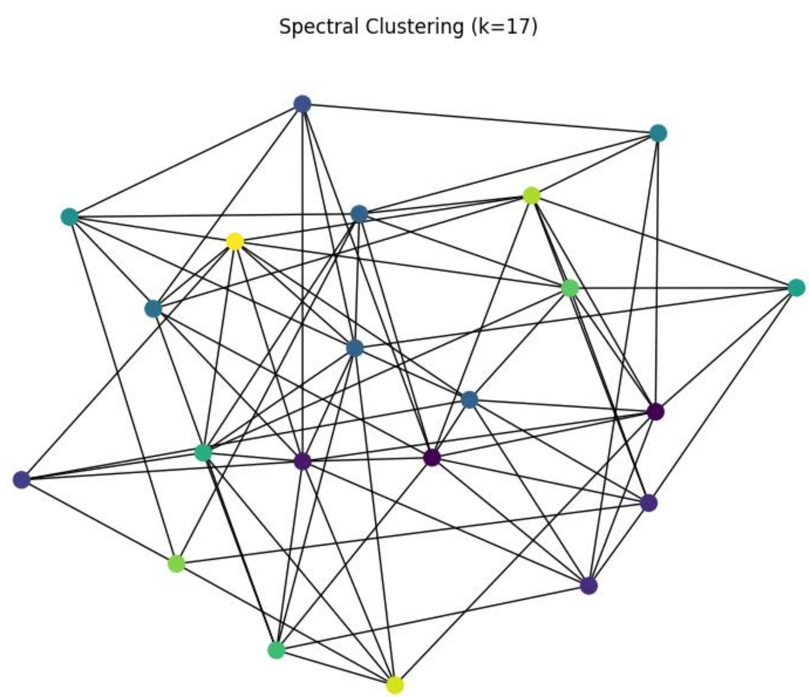

Fig 8 — Spectral clustering result for weak structure; effectively random

4.4 Fiedler Vector Bipartitioning

The Fiedler vector also fails to separate communities cleanly. ARI = −0.0154. The Fiedler approach can only produce two groups, and the weak signal makes even those two groups wrong.

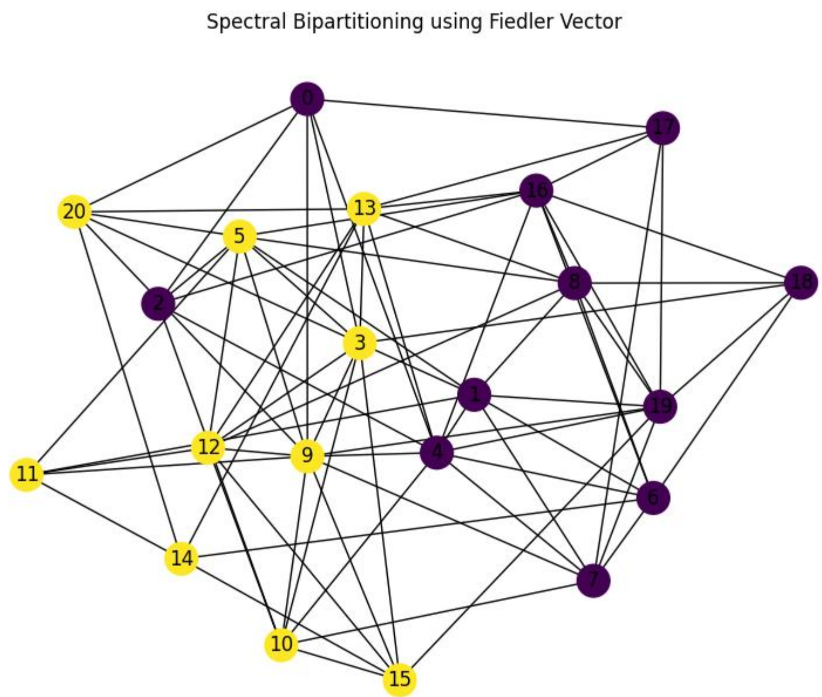

Fig 9 — Fiedler bipartitioning on weak-structure graph; communities are mixed

Observation

- ARI = −0.0154 for both methods — worse than a random assignment.
- The eigengap heuristic gives an unreliable community count.
- When intra and inter probabilities are close, the Laplacian cannot distinguish communities.
- This experiment shows the clear limitation of spectral methods on fuzzy structures.

5. Experiment 3 — Moderate Community Structure

Graph setup: 3 communities of 7 nodes each (21 nodes total). Intra-community probability = 0.70, Inter-community probability = 0.10. The difference is moderate — communities are distinguishable but not perfectly clean.

5.1 Graph Visualisation

The spring layout shows three loosely identifiable clusters. Ground truth colouring reveals three groups that are mostly separated with a few inter-community links.

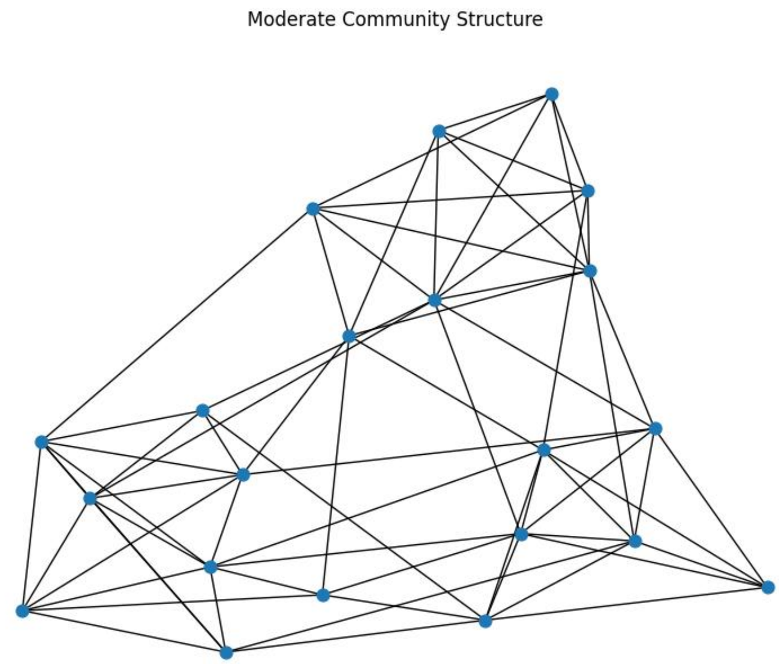

Fig 10a — Moderate-structure graph (no colouring)

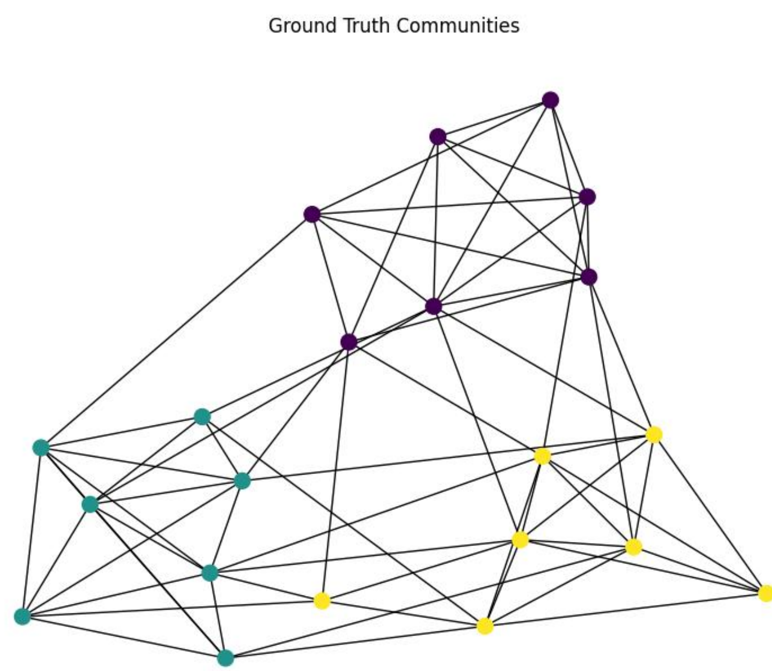

Fig 10b — Ground truth colouring; three communities visible

5.2 Eigengap Analysis

The gaps between eigenvalues are:

- Gap 1 = 1.6318 (ignored)
- Gap 2 = 0.8242
- Gap 3 = 1.4651 ← Largest effective gap
- Gap 4 = 0.7892 …

The largest gap (after ignoring Gap 1) occurs at position 3 in the gap array. Using the formula k = argmax(gaps[1:]) + 2, we get k = 1 + 2 = 3. The heuristic correctly identifies 3 communities.

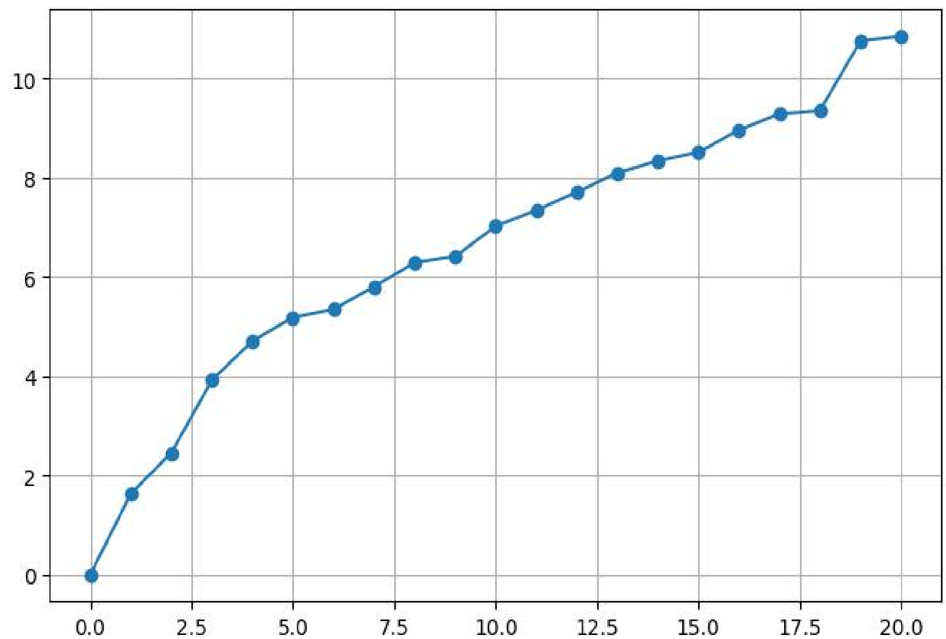

Fig 11 — Eigenvalue spectrum for moderate-structure graph (elbow visible around index 3)

5.3 Spectral Clustering Result

With k = 3, spectral clustering achieves ARI = 0.8533. Most nodes are correctly assigned; a few near community boundaries are misclassified.

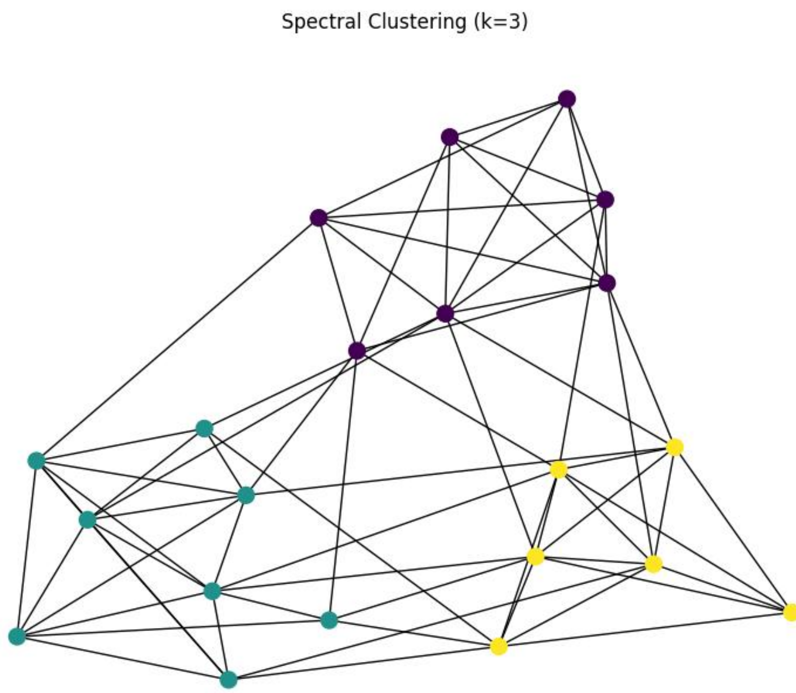

Fig 12 — Spectral clustering result (k = 3); good but not perfect recovery

Observation

- ARI = 0.8533 — good recovery but not perfect.
- The eigengap heuristic correctly identifies k = 3.
- Moderate community strength is enough for spectral methods to work well.
- A few nodes near the boundary between communities may be misassigned.

6. Final Summary

The table below compares all three experiments side by side.

Experiment	Intra-Prob	Inter-Prob	Actual Communities	Eigengap Estimate	ARI	Observation
Strong Structure	0.90	0.05	2	2	1.0000	Perfect recovery
Moderate Structure	0.70	0.10	3	3	0.8533	Good recovery
Weak Structure	0.50	0.35	3	Unreliable	-0.0154	Community overlap; failure

6.1 Key Takeaways

- Spectral clustering works best when communities are well separated (high intra, low inter probability).
- As inter-community probability increases, communities become harder to distinguish and ARI drops.
- The eigengap heuristic is reliable for strong/moderate structures but breaks down for weak structures.
- The Fiedler vector is a quick tool for two-community partitioning; for more communities, Spectral Clustering with KMeans is needed.
- The Stochastic Block Model is a great benchmark because the true community labels are always known.

6.2 Conclusion

Spectral methods for community detection rely on the mathematical structure of the Graph Laplacian. When communities are clearly defined, eigenvalues form distinct clusters and both the eigengap heuristic and spectral clustering perform perfectly. When communities blur together, the eigenvalue signal weakens and the algorithm struggles. The three experiments in this notebook provide a clear picture of both the power and the limitations of spectral community detection.